

Consistency Questionnaire Results

Q1. Missing physiological attributes

■ No inconsistencies found.

Q2. Multiple HR per ActorState

■ No inconsistencies found.

Q3. Multiple HRV per ActorState

■ No inconsistencies found.

Q4. Multiple RR per ActorState

■ No inconsistencies found.

Q5. Multiple SpO2 per ActorState

■ No inconsistencies found.

Q6. Multiple Drowsiness per ActorState

■ No inconsistencies found.

Q7. Missing high-level attributes

■ No inconsistencies found.

Q8. Multiple Fatigue per ActorState

■ No inconsistencies found.

Q9. Multiple Attention per ActorState

■ No inconsistencies found.

Q10. Multiple Unresponsiveness per ActorState

■ No inconsistencies found.

Q11. Multiple EyeState per ActorState

■ No inconsistencies found.

Q12. Multiple MouthState per ActorState

■ No inconsistencies found.

Q13. Multiple Age or Sex per ActorState

■ No inconsistencies found.

Q14. Multiple Accessories/Demographic/FaceCharacteristics per ActorState

■ 7 inconsistencies found.

```
Running Q14. Multiple Accessories/Demographic/FaceCharacteristics per ActorState...
■ Inconsistencies found (5 rows)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_ac1766d740514dbca1c
27f5260c477d9'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_0829f6d8e6fe477396b
e54cb422da3ea'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_35fc34e8ab7f4bdaa0a
55b860947e205'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_1109f38e03194099ab9
24e37fcd014da'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_d670ca63fab543c494d
c2c8f80e3ffde'),)
```

Q15-A. HR instances classified into more than one HR-level class

■ No inconsistencies found.

Q15-B. HRV instances classified into more than one HRV-level class

■ No inconsistencies found.

Q15-C. RR instances classified into more than one RR-level class

■ No inconsistencies found.

Q15-D. SpO2 instances classified into more than one SpO2-level class

■ No inconsistencies found.

Q15-E. Drowsiness instances classified into more than one Drowsiness-level class

■ No inconsistencies found.

Q16-A. HR without any HR-level classification

■ No inconsistencies found.

Q16-B. HRV without any HR-level classification

■ No inconsistencies found.

Q16-C. RR without any RR-level classification

■ No inconsistencies found.

Q16-D. SpO2 without any SpO2-level classification

■ No inconsistencies found.

Q16-E. Drowsiness without any Drowsiness-level classification

■ No inconsistencies found.

Q17-A. HR instances with >1 HR level

■ No inconsistencies found.

Q17-B. HRV instances with >1 HRV level

■ No inconsistencies found.

Q17-C. RR instances with >1 RR level

■ No inconsistencies found.

Q17-D. SpO2 instances with >1 SpO2 level

■ No inconsistencies found.

Q17-E. Drowsiness instances with >1 Drowsiness level

■ No inconsistencies found.

Q18. Violation "Sleep=> Inattentive & Unresponsive"

■ No inconsistencies found.

Q19. Violation "Drowsy & attentive=> at-risk unresponsiveness"

■ No inconsistencies found.

Q20. Violation "Sleep or unresponsiveness => eyes closed & mouth open"

■ No inconsistencies found.

Q21. Is any state missing its threshold profile?

■ No inconsistencies found.

Q22. Are there any threshold profiles not used by an Actor State?

■ 7 inconsistencies found.

```
Running Q22. Are there any threshold profiles not used by an Actor State?...
■ Inconsistencies found (17 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_young_female_cold'),)
```

Q23. Detect future categories accidentally left unmapped

■ No inconsistencies found.

Q24. Show if new data falls outside modeled thresholds.

■ No inconsistencies found.

Q25. Check if new states appear without corresponding GCI or SWRL effects.

■ No inconsistencies found.

Q26. Unexpected extensions

■ 7 inconsistencies found.

```
Running Q26. Unexpected extensions...
■ Inconsistencies found (2334 rows)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#Ontology'),)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdfLib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
```

Q27. Severe drowsiness (Level 9) must imply Imminent or Unresponsive

■ No inconsistencies found.

Q28. Moderate drowsiness (Level 5) must be classified as DrowsinessSuspected fatigue

■ No inconsistencies found.

Q29. Low drowsiness (Level 3) must be classified as Awake fatigue

■ No inconsistencies found.

Q30. Low drowsiness (Level 3) must imply Attentive attention

■ No inconsistencies found.

Q31. High drowsiness (Level 7 or 9) must imply Unresponsiveness or DrowsinessSuspected

■ No inconsistencies found.

Q32. Sleep fatigue must have both eye and mouth state defined

■ No inconsistencies found.

Q33. Awake fatigue must be Responsive and Attentive

■ No inconsistencies found.

Q34. High-risk states (Sleep or Unresponsive) must have at least one Label

■ 7 inconsistencies found.

Running Q34. High-risk states (Sleep or Unresponsive) must have at least one Label...

■ Inconsistencies found (10 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_0829f6d8e6fe477396b  
e54cb422da3ea'),)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_d670ca63fab543c494d  
c2c8f80e3ffde'),)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_0829f6d8e6fe477396b  
e54cb422da3ea'),)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_d670ca63fab543c494d  
c2c8f80e3ffde'),)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_0829f6d8e6fe477396b  
e54cb422da3ea'),)
```

Q35. Every Label must target at least one ActorState

■ 7 inconsistencies found.

Running Q35. Every Label must target at least one ActorState...

■ Inconsistencies found (25 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_ac1766d740514dbca1c  
27f5260c477d9'),)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_0829f6d8e6fe477396b  
e54cb422da3ea'),)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_35fc34e8ab7f4bdaa0a  
55b860947e205'),)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_1109f38e03194099ab9  
24e37fcd014da'),)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_d670ca63fab543c494d  
c2c8f80e3ffde'),)
```